

Functional and differential equations

— 4

Linear equations, systems and boundary value problems

Homework

13 сентября 2026 года

Theoretical minimum. The solution space of a linear homogeneous equation of order n has dimension n , the Wronskian obeys Abel's identity, and the inhomogeneous equation is solved by variation of parameters. With constant coefficients everything is the matrix exponential: the Jordan form dictates the shape $t^j e^{\lambda t}$ and the real parts of the eigenvalues dictate the behaviour at infinity. With a variable matrix the exponential is replaced by the Peano series, which collapses to $\exp \int A$ exactly when the matrices commute. Boundary value problems behave like linear algebra rather than like initial value problems: solvability is a Fredholm alternative, uniqueness means that the homogeneous problem has only the zero solution, and the eigenvalues of the Sturm–Liouville problem are the obstruction. Green's function turns the boundary value problem into an integral operator, which is how existence is proved.

1. Compute e^{At} for a 2×2 matrix in all three cases (distinct real, complex conjugate, repeated eigenvalue), and describe the corresponding phase portraits.
2. Prove that all solutions of $\dot{x} = A(t)x$ with $\int_0^\infty \|A(t)\| dt < \infty$ have finite limits as $t \rightarrow \infty$, and that the limit map is invertible.
3. Prove that $\det \Phi(t) = \det \Phi(0) \exp \int_0^t \operatorname{tr} A(s) ds$ for a fundamental matrix Φ of $\dot{x} = A(t)x$ (Liouville's formula), and deduce Abel's identity for a scalar equation of order two.

4. Construct Green's function for $y'' = g$ with $y(0) = y(1) = 0$, prove that it is symmetric, and use it to show that the boundary value problem has a unique solution for every continuous g .
5. Prove the Fredholm alternative for the problem $y'' + \lambda y = g$, $y(0) = y(\pi) = 0$: it is solvable for every g when λ is not an eigenvalue, and solvable only for g orthogonal to the eigenfunction when it is.
6. Let q be continuous and non-negative. Prove that the problem $y'' = q(x)y$, $y(0) = y(1) = 0$ has only the zero solution, and give an example with q of variable sign where this fails.
7. Prove that for the periodic problem $\dot{x} = A(t)x$ with $A(t+T) = A(t)$ the monodromy matrix $\Phi(T)$ determines stability, and state Floquet's theorem.
8. Solve $x^2 y'' + xy' + (x^2 - \nu^2)y = 0$ near $x = 0$ by a power series (Bessel's equation), and determine for which ν there is a solution bounded at the origin.
9. Prove that the eigenvalues of the Sturm–Liouville problem $-(py')' + qy = \lambda wy$ with positive p, w and Dirichlet conditions are real, simple and tend to infinity, and that the eigenfunctions are orthogonal with weight w .
10. Let $y'' + q(x)y = 0$ with a continuous q have a solution with two zeros in $[a, b]$. Prove Lyapunov's inequality $\int_a^b |q| > \frac{4}{b-a}$, and show that the constant 4 cannot be improved.